林下之网

The Web Beneath the Woods



树木的地下联系，哪些已有证据?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

One of ecology's most romantic ideas grew out of plastic bags. In a Canadian forest, researchers enclosed paper birch and Douglas fir seedlings in them, then bathed each species in carbon dioxide laced with a different tracer. Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight surplus, larger in deeper shade.

Most terrestrial plants harbour such fungi, whose filaments forage for water and nutrients including nitrogen, swapping them for plant sugars. One fungus can knit plants into a shared network. The journal Nature unveiled the study on 7 August 1997, its cover offering a catchy coinage: the "wood-wide web".

The metaphor proved irresistible. Between 2015 and 2021, a forester's bestseller, an acclaimed novel, and an ecologist's memoir told readers that trees converse, warn one another, and, as "mother trees", nurture their kin. Yet the 1997 trees were seedlings, not elderly mothers.

In February 2023, three researchers combed through the field evidence. They judged it too sparse, inconsistent, or ambiguous to demonstrate that shared networks are widespread. In 26 field studies, networks helped and hindered seedlings about equally often, with neutral effects predominating. No peer-reviewed field study showed mature trees favouring their own offspring. Unsupported assertions in later papers had doubled over 25 years, with citations skewed towards positive results.

None of this disproves the networks' existence. In January 2025, defenders faulted the critique's methods. The critics concede that carbon and nutrients can migrate between trees underground. It is evidence unearthed in real forests, not the allure of a family saga, that must settle how much the networks convey, and whether it matters.

中文译文

生态学中一个格外浪漫的观念，竟起源于塑料袋。在加拿大的一片森林里，研究人员把纸桦和花旗松的幼苗罩进袋中，分别让两种幼苗接触掺有不同示踪物的二氧化碳。碳在两种树之间双向转移，似乎经由连接根系的真菌传递。不过，花旗松最终获得了少量净盈余，在遮荫更深的地方，盈余也更大。

大多数陆生植物都与这类真菌共存。真菌的菌丝寻找水分和包括氮在内的养分，再拿这些资源换取植物的糖分。一个真菌就能将多株植物连成共享网络。《自然》于1997年8月7日发表这项研究，封面还用上了一个好记的新词：“林木万维网”。

这个比喻实在太有吸引力。2015年至2021年间，一位护林人的畅销书、一部广受好评的小说和一位生态学家的回忆录，都告诉读者：树木能够交谈、相互示警，还会像“母树”一样养育亲族。可1997年实验中的树只是幼苗，并不是年迈的母亲。

2023年2月，三位研究者系统梳理了野外证据。他们认为，这些证据或太少、或互不一致、或含糊不清，不足以证明共享网络普遍存在。在26项野外研究中，帮助与妨碍幼苗的结果出现得大致同样频繁，而最多见的是没有明显影响。没有一项经同行评审的野外研究表明，成年树木会偏爱自己的后代。后续论文中缺乏依据的断言在25年间翻了一倍，引用也偏向正面结果。

这些批评并不能否定网络的存在。2025年1月，网络说的拥护者质疑了那篇批评文章的方法；批评者也承认，碳和养分能够在树木之间经地下迁移。网络究竟传输了多少、这种传输在生态上有多大作用，必须由真实森林中的证据来判定，不能取决于一段“树木家族传奇”有多动人。

“无明显影响最多”与“帮助、妨碍的次数接近”并不矛盾；复核未证明网络不存在。

核心词汇

先记这22项。连着下方的原文短语一起回想。

ecology n.
生态学
One of ecology's most romantic ideas

seedling n.
幼苗
paper birch and Douglas fir seedlings

tracer n.
示踪物
a different tracer

apparently adv.
似乎；看来
apparently via fungi

surplus n.
盈余
a slight surplus

terrestrial adj.
陆生的
Most terrestrial plants

filament n.
丝状体；此处指菌丝
whose filaments forage for water

nutrient n.
养分
water and nutrients including nitrogen

metaphor n.
比喻
The metaphor proved irresistible

nurture v.
养育；培育
nurture their kin

comb through 动词短语
仔细梳理；彻底搜查
combed through the field evidence

sparse adj.
稀少的
too sparse, inconsistent, or ambiguous

inconsistent adj.
互不一致的
inconsistent, or ambiguous

ambiguous adj.
含糊不清的
inconsistent, or ambiguous

demonstrate v.
证明
to demonstrate that shared networks

widespread adj.
普遍存在的
shared networks are widespread

neutral adj.
无明显正负影响的
neutral effects predominating

peer-reviewed adj.
经同行评审的
No peer-reviewed field study

assertion n.
断言
Unsupported assertions

citation n.
引文；引用
with citations skewed

concede v.
承认（先前不愿承认的事）
The critics concede

convey v.
传输；输送
how much the networks convey

文中词汇与搭配

补充词汇共50项。保留每个表达在本文中的意思。

romantic adj.
富有浪漫色彩的

grow out of 动词短语
源自；由……发展而来

enclose v.
把……围住；罩住

in prep.
在……里面

bathe v.
使浸在；使充分接触

be laced with 动词短语
掺有

shuttle v.
往返移动

fungus n.
真菌

link v.
连接

slight adj.
少量的；轻微的

shade n.
遮荫

harbour v.
容纳；携带

forage for 动词短语
搜寻（水或养分）

nitrogen n.
氮

swap v.
交换

knit v.
联结；编织

into prep.
进入；连成

shared network n.
共享网络

unveil v.
公布；发表

catchy adj.
好记而吸引人的

coinage n.
新造词

irresistible adj.
难以抗拒的

bestseller n.
畅销书

acclaimed adj.
广受赞誉的

memoir n.
回忆录

文中词汇与搭配 续

补充词汇共50项。保留每个表达在本文中的意思。

converse v.
交谈

warn one another 动词短语
互相警告

kin n.
亲族

elderly adj.
年迈的

hinder v.
妨碍

predominate v.
占多数

mature adj.
成熟的；成年的

favour v.
偏爱

offspring n.
后代

unsupported adj.
缺乏依据的

double v.
翻倍

skew v.
使偏向

towards prep.
朝；偏向

positive result n.
正面结果

disprove v.
证明……不成立

existence n.
存在

defender n.
拥护者

fault v.
指责；挑出问题

critique n.
批评性分析

critic n.
批评者

migrate v.
迁移

unearth v.
发掘

allure n.
吸引力

saga n.
传奇故事

matter v.
有重要意义

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Carbon shuttled both ways, apparently via fungi linking the trees' roots, yet the firs ran a slight _____, larger in deeper shade.

02 The _____ proved irresistible.

03 In 26 field studies, networks helped and hindered seedlings about equally often, with _____ effects predominating.

04 The critics _____ that carbon and nutrients can migrate between trees underground.

文章主线

- 发现 1997年，示踪实验发现纸桦和花旗松幼苗双向交换碳，似乎经连接根系的真菌传递。
- 走红 真菌以水和养分换植物糖分，也能连接多株植物；“林木万维网”走红后，被演绎成母树哺育亲族。
- 复核 2023年复核认为网络普遍存在的证据不足；26项野外研究以无明显影响为主，帮助和阻碍幼苗的次数相近。
- 偏差 复核未找到母树偏爱亲生后代的同行评审野外证据；论文偏引正面结果，无依据断言在25年间翻倍。
- 争议 2025年拥护者质疑复核方法；批评者也承认树间有地下物质转移，但网络输送多少、生态作用多大仍待实证。

参考答案　1 surplus 2 metaphor 3 neutral 4 concede